ModelIt!

Student Activity Pack

Climate Feedback Loops: Why Small Changes Get Big

How CO_2, Temperature, Ice, and Reflectivity Create a Runaway System

Grade 08L2 | NGSS MS-ESS2-6, MS-ESS3-5 | Stability and Change

STUDENT NAME _____ DATE _____

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Table of Contents

ModelIt!

1 Pre-Assessment 3

2 Scoring Rubric 4

3 Component Analysis 5

4 Model Recording Page 6

5 Simulation Observations 7

6 Research & Extend 8

7 Reflection Questions 9

8 STEM Challenge 10

9 Post-Assessment 11

1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1 What is the greenhouse effect, and how does it relate to global warming?

2 What do you think happens when Arctic ice melts?

3 Have you heard of a 'feedback loop'? What do you think it means?

4 Draw a diagram showing how you think CO2 in the atmosphere affects Earth's temperature.

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

CO2 Emissions Rate | Atmospheric CO2 Level | Global Temperature | Ice Cover | Albedo (Reflectivity)

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Pre-Industrial Baseline

Low CO_2 emissions, natural levels

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Current Emissions

Modern emission levels

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Runaway Feedback

Very high emissions sustained for 50 years

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Permafrost Thaw — Frozen ground that releases methane — an even stronger greenhouse gas — when it melts

Ocean Heat Absorption — Oceans absorb heat, slowing atmospheric warming but causing marine impacts

Cloud Cover Changes — Warming changes cloud patterns, which can either reflect or trap more heat

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why do small increases in CO_2 lead to such large changes in global temperature? Use evidence from your simulation to support your answers.

1. **Why is the ice-albedo feedback loop called a 'positive' feedback even though the effects are negative?**

2. **How does the concept of a tipping point make climate change different from gradual temperature change?**

3. **Why is it harder to reverse a feedback loop than to stop it before it starts?**

4. **What does your model reveal about the difference between 'linear thinking' and 'systems thinking'?**

5. **How might understanding feedback loops change the way governments approach climate policy?**

8 STEM Challenge: Design a Climate Action Plan

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YOUR ENGINEERING MISSION

Design an evidence-based climate action plan for your city that targets the feedback loop at its most effective intervention point, using data from your model.

THE SCENARIO

Your city council has asked your team to propose a climate action plan. They want to know: where in the CO2-temperature-ice feedback loop can we intervene most effectively, and what measurable goals should we set?

Engineering Design Thinking

1. WHAT SPECIFIC EMISSION REDUCTIONS WILL YOUR PLAN TARGET?
2. HOW WILL YOU ADDRESS THE FEEDBACK LOOP'S SELF-REINFORCING NATURE?
3. WHAT TIMELINE DOES YOUR MODEL SUGGEST FOR ACHIEVING STABILIZATION?
4. HOW WILL YOU MEASURE PROGRESS AND COMMUNICATE RESULTS TO THE PUBLIC?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Climate Scientists and Atmospheric Researchers study feedback loops in Earth's climate system and develop models to predict future changes. They earn $70,000-$120,000/year and work at universities, government agencies, and research institutions.

9 Post-Assessment

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Show What You Learned!

Multiple Choice: Now that you have completed the lesson, circle the best answer for each question. Compare your answers to your pre-assessment — did your thinking change?

1 Your model shows that at 90% CO_2 Emissions Rate, Global Temperature increases exponentially rather than linearly. Which model feature explains this acceleration?

- **A.** CO_2 directly heats the atmosphere without any intermediate steps
- **B.** The positive feedback loop where warming reduces ice, which lowers albedo, which causes more warming, amplifies the initial CO_2 effect
- **C.** Higher CO_2 levels cause the sun to emit more radiation
- **D.** Temperature increases always follow an exponential pattern regardless of the cause

2 In the model, Albedo has a NEGATIVE relationship with Global Temperature. A student is confused because lower albedo causes higher temperature, which seems like it should be positive. Which explanation correctly resolves this?

- **A.** The relationship is mislabeled in the model and should be positive
- **B.** Higher albedo reflects more sunlight away from Earth, which cools the surface. The negative sign means the two values move in opposite directions.
- **C.** Albedo and temperature are not actually related in the model
- **D.** The negative sign indicates that albedo destroys thermal energy

3 The model shows that even after CO_2 Emissions Rate is reduced to 0%, Global Temperature continues rising for several decades. What does this evidence reveal about the climate system?

- **A.** The model is flawed because temperature should immediately decrease when emissions stop
- **B.** The feedback loop has its own momentum. Reduced ice cover continues absorbing extra sunlight, maintaining warming even without new CO_2 input
- **C.** CO_2 stays in the atmosphere forever and never breaks down
- **D.** Reducing emissions has no effect on climate change

4 A student claims that the ice-albedo feedback loop is a 'positive feedback' that produces positive results for Earth. Using the model, which response best corrects this misunderstanding?

- **A.** The student is correct because positive feedback always produces beneficial outcomes
- **B.** In systems science, 'positive' means the feedback amplifies the original change, not that the outcome is good. The ice-albedo loop amplifies warming, which has harmful consequences.
- **C.** The feedback is actually negative because it produces harmful effects
- **D.** The terms 'positive' and 'negative' have no specific meaning in feedback systems

5 **Using your model, you must advise a city council on climate action. The model shows that intervening at CO_2 Emissions Rate has the largest effect on all downstream components. Why is this the most effective intervention point?**

A. Because CO_2 is the only component that humans can control

B. Because CO_2 Emissions Rate is the external input that drives the entire feedback loop. Reducing it lowers Atmospheric CO_2, which slows temperature rise, which preserves ice, which maintains albedo.

C. Because reducing CO_2 immediately reverses all warming that has already occurred

D. Because CO_2 is the most expensive component to address